

Beyond Spatiotemporal Substantialism: A Case for a Relational Ontology of Information

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Abstract

Contemporary metaphysics of physics is still largely committed to a spatiotemporal substantialism inherited from general relativity in which spacetime is hypostatized as a primitive fundamental entity. This paper proposes an ontological shift to informational monism, based on the scheme of Ontic Structural Realism (OSR). We argue that relational information is the irreducible ontic substrate of reality, and that space and time are macroscopic epiphenomena, the foundational principles of which can be epistemologically deconstructed. In this context, the phenomenological flow of time is rephrased as the sequential actualization of states in the informational network, and gravitation loses its role as an efficient cause to become a syntactic constraint on the bandwidth of these exchanges. The global informational volume is strictly conserved. We show that the antinomy between the fundamental reversibility of the quantum evolution and the irreversibility of the arrow of time is resolved by replacing the classical concept of "entropic degradation" with the concept of "topological delocalization".

Keywords: Informational Monism, Ontic Structural Realism, Emergent Gravity, Thermal Time, Philosophy of Physics, Entropy.

Epigraphe

'It from bit. Or conversely, every 'it' - every particle, every field, even the space-time continuum itself - derives its function, its meaning, its very existence from the apparatus-definite answers to yes-or-no questions, binary choices, bits." John Archibald Wheeler (1990)

“Time is not a fundamental category, it is a derived concept for subsystems. The whole universe does not evolve.” — Carlo Rovelli (1993)

Prologue: A Brief Genealogy of the Substantialism-Relationalism Controversy

Before we make our positive proposal it is useful to recall the conceptual landscape we inherit: The opposition of Newton's absolute space (duration as a 'sensorium of God ') and Leibniz's relationalism (space as 'order of coexistences') was never conclusively resolved - only displaced. Kant attempted to do this synthesis through his transcendental idealism (space and time as a priori forms of intuition). Ironically, it was Einstein's general relativity that revived the substantialist reading: the metric field is a dynamical entity with its own degrees of freedom.

Since then, this consensus has been tested by three major crises:

- **The problem of time in quantum gravity** (Wheeler-DeWitt equation). The Hamiltonian constraint leads to a static wavefunction of the universe, frozen dynamics.
- **Black hole thermodynamics**: The Bekenstein-Hawking entropy indicates information is proportional to area, not volume – a non-spatiotemporal locus.
- **Holographic principle** ('t Hooft, Susskind): the content of a volume can be described by degrees of freedom on its boundary, against substantialist intuition.

Our informational monism is not some exotic speculation, but the sober attempt to take these three crises at their ontological word.

1. Introduction: The Impasse of Geometric Substantialism

(Terminological warning to the reader: “information” in this essay is not to be taken in the New Age or popular spiritual sense. It is not “consciousness” or “vibration” or “energy” or any mystical substance. Information is strictly defined as relational difference — a formal, mathematical, and physically grounded concept. Reading panpsychism, spiritualism, or quantum woo into this framework is a category error.)

Modern physics was a debate between Leibniz's relationalism (space as order of coexistences) and Newton's substantialism (absolute space). In the three centuries of physics, the operational triumphs of Newtonian mechanics reigned; and, even though Einstein's general relativity united space and time in a dynamic fabric, it also preserved a substantialist ontology: spacetime acts on matter, and is acted upon by matter.

But the stubborn inability of gravity quantization and the paradoxes of black hole thermodynamics (notably the information paradox) indicate that spacetime is not the ultimate substrate of reality. To get round this, it is necessary to suspend the usual geometric analogies, and to go back to first principles. This paper draws from Wheeler's 'It from bit' (1990) and subsequent work in the philosophy of information to promote a radical metaphysical thesis: that the basic building blocks of nature are not point particles, continuous fields, or the spatiotemporal continuum itself, but pure informational relations. Space and time are the macroscopic illusion of the density of underlying quantum entanglements . Objects do not inhabit space and time .

2. A Distinction That Matters: Shannon Information vs. Ontic Information

We must clear up a common equivocation before going any further. Shannon information ($H = -\sum p_i \log p_i$) is an epistemic measure: it measures an observer's uncertainty about the state of a system, conditioned on the probability distribution. That is a matter of knowledge.

Ontic information (what we call "informational substance") is different: it is the absolute relational difference that constitutes a physical state, independent of any observer. When realized as , say , a spin up or down , classical bit (0 or 1) is ontic information . The same bit has one Shannon bit of uncertainty for a person who knows the value of the same bit.

Our thesis is one of ontic information alone. Ontic information is subject to the conservation principle (axiom 3). In particular, Shannon entropy can locally increase (i.e. ignorance can increase) while ontic information is strictly conserved - precisely because delocalization makes ontically structured information epistemically inaccessible correlations. This, in particular, dispels a common misconception: we are not saying that "information is physical" in a trivial sense (Landauer), but that physical reality is the structure of ontic differences.

3. Axiomatic of Informational Monism

In order to isolate the core conflict in modern physics (spatiotemporal continuity versus quantum discretization) our framework is based on five metaphysical postulates that re-define the nature of substance and neutralize the classical ontological division between matter and space:

- **Ontic Primacy of Information.** In the spirit of Ontic Structural Realism (Ladyman & Ross, 2007), information is not an epistemological category measuring our ignorance, but the ultimate ontological substance. Information here is not understood as a property of some pre-existing

substratum (matter or field) but as pure relational difference that constitutes the very texture of reality. There is no information of something; there are differences that make a difference and these differences are all there is. Support of information is information in its relational form.

- **Interaction as a Logical Operator:** Classical Efficient Causality Replaced by Formal and Computational Causality. A physical interaction is simply the evaluation of a relational Hamiltonian changing the state of two subsystems. This is not an externalized computation (by a Laplacean demon or a cosmic computer), but an immanent self-evaluation of the informational network. A physical interaction is a local realization of a logical constraint inscribed in the relational structure. By “computation” in this context we simply mean that one relational state entails another according to determinate rules—without presupposing any external processor.
- **Absolute Conservation of Being (as a Regulative Principle) :** The global informational content of the Universe is strictly conserved, in accordance with the unitary evolution of quantum mechanics. We assume as a metaphysical principle the conservation of total information. In the unitary framework of quantum mechanics this means taking the whole universe as a pure state ($S(\rho_{\text{universe}}) = 0$). This assumption is, however, empirically undecidable for any internal observer. It is a regulator of ontological discourse: any local increase of entropy must be compensated by an informational depletion elsewhere or by a delocalization.
- **Temporal Emergentism :** Time is not a fundamental dimension It is the statistical epiphenomenon of the logical sequence of local informational transformations, which results from the incompleteness of the observer.
- **Gravitation as Entropic Syntax :** Gravity is not a fundamental attraction force but an organizational law that governs the routing of exchanges over the relational graph.

4. A Minimal Formal Characterization

To avoid the framework being entirely metaphorical, we present a minimal formal skeleton.

Let $G = (N, E)$ be the directed graph of the basic informational network:

- N = nodes (not “things” but loci of relations; indistinguishable except by their place in the graph).
- E = edges (binary information differences). Each edge is labelled by a value from a finite set Σ (the alphabet of elementary differences).

The state of the universe is a function $\phi: E \rightarrow \Sigma$

The dynamics (without time) is a set of rewriting rules R : patterns of edge values replaced by other patterns. The universe “evolves” through the application of R over the graph simultaneously. Two derived ideas:

- **Local information density** $\rho(i)$ for node i : the (Shannon) entropy of the distribution of edge values incident on i , computed relative to the global state.
- Relational bandwidth $B(i,j)$: number of distinct edges between i and j (multigraphs allowed).

The time measured by a local subsystem $S \subset N$ is then defined as:

$$\tau_S = f(|\{ t \in \mathbb{N} : \varphi_t|_S \neq \varphi_{t-1}|_S \}|)$$

where t indexes logical steps (rather than temporal duration), and f is a monotonic function based on S 's internal clock (e.g., frequency of state changes). This formalisation makes it plain: time is the count of local changes, not a fundamental parameter.

Gravity as a syntactic constraint: The rewriting rules R are not arbitrary, but minimize the total relational bandwidth needed to propagate differences across the graph. The continuous limit of this minimisation is Einstein's equations. We state this as a conjecture, not proven here, but empirically plausible given Verlinde's derivation of Newton's law from entropic reasoning.

5. The State of Time: From Absolute Flow to Relational Actualization

The phenomenological experience of the passage of time is at odds with the static block-universe of relativity. The conflict is solved by our relational ontology with the notion of thermal time (Rovelli, 1993).

If the global information is invariant, then time does not have an objective existence at the level of the fundamental graph of the Universe. "Becoming" can only come out of the partial point of view of a subsystem in interaction with its environment. The perceived speed of time is the hard metric of how often this subsystem updates its relative data. Relativistic time dilation thus becomes formally legible: a system deprived of the ability to interact (either by virtue of its speed approaching that of light or by virtue of its submersion in a field of high informational density) drains its computational power. Mechanically, it runs out of its relational bandwidth to “generate” local time.

A classical objection is to say that a “logical sequencing” already presupposes a temporal order. This objection confuses the phenomenal order with the metaphysical order. Logical order (for example: a formal cause comes before its

effect in the order of definition, not in time) implies no duration. But when a subsystem is incomplete, it must iterate its informational updates, and this is how temporal experience comes about. The ``passage" that is felt is the trace, in the local structure, of the global constraint of coherence of the network.

6. Three Competing Perspectives and Our Response

To sharpen up our contribution we contrast it with three prominent contemporary views:

Position	Proponent	Core Thesis	Our Response
Timeless relationalism	Barbour (1999)	Time is an illusion; reality is a "Platonia" of static configurations.	Agrees time is not fundamental, but retains a phenomenological residue unaccounted for: why is there an ordered experience of succession? Our emergence via incomplete updates explains <i>why</i> the illusion occurs, not just that it occurs.
Real time with becoming	Smolin (2013)	Time is real, fundamental, and precedes laws.	Empirically plausible but metaphysically costly. Smolin's view struggles to reconcile with general relativity's block universe. Our view keeps the block's formalism while explaining why inhabitants experience flow.
Primitive temporal direction	Maudlin (2007)	The arrow of time is a primitive, irreducible fact.	This is a stipulation, not an explanation. Our framework <i>derives</i> the arrow from delocalization without primitive postulates.

Thus, informational monism occupies a unique niche: it admits the formalism of relativity and quantum mechanics while explaining why time appears as it does - - something rival views either deny (Barbour) or postulate (Maudlin).

7. The Great Observer (God) as a logical limit

Our approach, in which time is an epistemic consequence of incomplete information, naturally leads to the question of what an omniscient observer, e.g. God, or a Laplacean demon, who knew the exact state of the entire universe, would experience.

Three possible answers must be distinguished, with different theological or metaphysical implications.

First reply (orthodox relationalism): the Great Observer has no time. If time is simply the measure of local informational incompleteness, then a being possessing complete information about the universal state, all nodes, all edges, all assignments $\varphi: E \rightarrow \Sigma$, experiences no succession. For such a being the universe

is a static, eternal structure. All logical steps exist simultaneously in the order of definition, not in the order of duration. This entity does not "wait" for anything. It doesn't "remember." It doesn't "anticipate." It just sees.

It is the classical block universe from the perspective of God, timeless, complete, and perfectly transparent. In this respect our metaphysics is compatible with a neo-Einsteinian theology: God does not exist in time, God is the name we give to the total informational structure itself. Second answer (process theology) : The Great Observer observes . This creates time .

An alternative, more radical interpretation: omniscience is not a static snapshot but an infinite act of simultaneous comprehension. If God must preserve the coherence of the informational network — if divine knowledge is not a passive reflection but an active synthesis — then the Great Observer is not external to the network but its maximal node. This node contains all relations and thus has the maximum possible relational bandwidth. And such a node, far from being timeless, would experience all possible temporal perspectives simultaneously — not a single linear time, but a superposition of all local times.

This resonates with certain process theologies (Whitehead, Hartshorne) where God is not the unmoved mover but the fellow sufferer who understands. In our language: God is the node with local time equal to the integral over all local times - an experience we cannot represent but can describe formally.

Third answer (skeptical): the Great Observer is a meaningless notion.

A more deflationary response: the concept of "omniscience" is incoherent in our framework. Information is relationally defined. There is no view from nowhere (Nagel). The observer is always (never outside) a node of the network. A being who knew the whole network would have to be the whole network itself -- but the network knows nothing. A knower is a subsystem . A knower is required for knowledge. The "God as observer" is therefore a grammatical illusion, not a possible being.

Our position: We accept the first answer as the logical consequence of the formalism, but we observe that the second and third are not incoherent, they simply choose different interpretations of "omniscience". What is important for our metaphysics is that none of these answers reintroduces a Newtonian absolute time. Not even the Great Observer, if it exists, lives in the time of physics textbooks. It is either nothing (static structure) or everything (superposition of local times) or the concept crumbles. Theological consequence If one wants to say anything about God, the only formulation that is consistent with this metaphysics is:

God is not the creator of the informational network, but the network in its total actuality, the complete graph of all differences, without lack, and therefore without time.

Very close to Spinoza's Deus sive Natura (God or Nature) and some readings of Einstein's 'cosmic religion'. This is not a personal God, but it is a coherent ontological anchor for the term 'divine'.

8. Gravity as a Syntactic Constraint on Organization

What is the metaphysical status of gravity if it is not a mechanical interaction in the sense of the Standard Model? Our monism, by philosophically transposing the model of entropic gravity (Verlinde, 2011), claims that gravity belongs to a formal cause only.

A macroscopic mass must be reconceptualized as a zone of ontological saturation: a hyper-dense node of the informational graph, exhibiting a massive gradient of density of states. Gravity is just the statistical effect that pushes subsystems to maximize their entanglement entropy by moving to such hyper-connected regions. The geometric curvature phenomena described by Einstein are the large-scale projected shadows of this algorithmic constraint which seeks to optimize the processing of fundamental information.

9. Cosmological Implications The Direction of Cosmic Time

Our framework predicts explicitly for the large scale arrow of time: the universe as a whole has no arrow. Time is local only. If the entanglement structures of different regions allow, they can have statistically opposite thermal arrows.

This is not a bug, but a feature. It forecasts:

- Early universe: low entropy (high local information structure, low delocalization)
- Cosmic expansion: not a metric expansion-in-time, but the increase in the diameter of the graph as new nodes and edges are "folded in" from previously inaccessible correlations.
- Dark energy: possible artefact of the information cost of maintaining global coherence as local subsystems become increasingly delocalised relative to each other. The accelerated expansion corresponds to saturation of relational bandwidth.

These are speculative but empirically accessible: they predict specific corrections to the Friedmann equations at high redshift (a potential signature in future surveys like Euclid or Roman). This we present not as a result but as a research programme.

10. An Aside on Hidden Variables: Bohm and 't Hooft

Our framework is naturally suited to comparison with hidden variable approaches, in particular Bohmian mechanics and 't Hooft's cellular automaton interpretation.

Similarities with 't Hooft (2016). Gerard 't Hooft has proposed that quantum mechanics is a coarse grained description of an underlying deterministic cellular automaton. Our informational graph is structurally similar. The rewriting rules R are deterministic at the basic level, and quantum indeterminacy arises from relational incompleteness. The main difference is that 't Hooft retains a primitive notion of time (the automaton step) while we eliminate it altogether, replacing it with logical succession.

Bohm (1952) divergences. Bohmian mechanics introduces a non-local pilot wave and real particle positions. Our framework does not have particles, does not have a wave function as a separate entity, and does not have a preferred foliation of spacetime. Bohm's ontology is dual (particles + wave); ours is monist (information only). Bohm's non-locality is kinematic, ours is structural.

What we borrow: The idea (from 't Hooft) that quantum mechanics and determinism can co-exist. From Bohm the insight that non-locality is not a bug but a feature of relational ontologies. But we go further. Particles are not fundamental, nor is time.

A testable difference: We predict in our framework that any hidden variable theory needing a primitive temporal parameter cannot be successful (Bohm's time is Newtonian). 't Hooft's cellular time might be compatible but only if reinterpreted as logical succession - which is empirically indistinguishable from ordinary time in present experiments. The divergence appears only at the Planck scale, where our framework predicts discrete, but not temporal graph dynamics.

11. Topological Delocalization: The Resolution of the Entropic Conflict

One of the main epistemological issues is to reconcile the strict conservation of quantum information with the irreversibility of the Second Law of Thermodynamics. How can disorder increase without loss of global information?

Our framework offers a purely topological interpretation, with no theoretical compromises. If information can not be destroyed, then the increase in entropy is just a growing relational delocalization. When a structured system (low entropy) dissipates its initial information is not erased, but rather migrates from macroscopic degrees of freedom to inaccessible microscopic degrees of freedom, becoming entangled with the environment. The universe doesn't decay, it

entangles itself. Local order becomes epistemic opacity, but global ontological transparency is preserved.

11. Objections and Response

Objection 1 (testability). Gravity as a syntactic constraint is unfalsifiable. This removes the framework from science.

Reply: This objection applies a Popperian criterion to a scientifically-oriented metaphysics. We do not want to replace general relativity, but to propose an ontological interpretation that dissolves its internal aporias (singularities, the information paradox). Indirect testability is by coherence with existing theories and the ability to make new qualitative predictions (e.g. about black hole thermodynamics).

Objection 2 (support of information) Relational information presupposes relata different from relations. A node in a graph is not reducible to its edges.

Reply: This is exactly the substantialist assumption, which we deny. In a pure and simple relational framework (radical OSR) relata are simply nodes defined solely by their relations. There is no substratum underneath: the network is all there is. The distinction between node and edge is a convenient abstraction, not an ontological one.

Objection 3 (the Great Observer objection). You say time is the consequence of the observer's incompleteness. But if an omniscient observer, God or a Laplacian demon, had perfect knowledge of the whole informational network, would it have no experience of time at all? And if so, does not that make time an illusion and your framework idealist?

Answer This objection confuses two different claims: (a) that time is an illusion only; (b) that time is a genuine relational property for finite systems. We support (b) not (a). The Great Observer, if it existed, would indeed have no temporal experience -- but this is because the Great Observer is not a subsystem, and therefore not an observer in the informational sense. By definition, an observer is a node that has incomplete access to the total network. The fact that a hypothetical complete knower would not see time does not imply that time is illusory for us. It just implies that time is perspectival. This is no more problematic than saying that color does not exist for a being with no photoreceptors, but color is real for trichromatic primates. Finite observers have real time. The Great Observer is a limiting case, not a contradiction.

Objection 4 (unitarity of the universe): The hypothesis $S(\rho_{\text{universe}})=0$ is strong and arguably undecidable. Can you build a metaphysics on an undecidable proposition?

Answer: Yes, if the proposition is a regulative principle (in the Kantian sense), not an empirical hypothesis. It poses the question: any local increase of entropy should be interpreted as delocalization, not destruction. The alternative, a fundamentally mixed state, would mean that information can be truly lost, which is in contradiction with the unitary core of quantum mechanics. Therefore, our choice is not arbitrary but theoretically driven.

12. How does Informational Monism Change Things? Beyond Physics

If the preceding arguments are right, the metaphysical picture of the world is radically changed. But metaphysics is not physics. It also influences how we position ourselves in reality.

Our framework yields three existential corollaries (tentatively proposed, not deduced):

The self as a local information center. ‘Consciousness’ or ‘self’ is a metastable pattern of informational differences that preserve coherence in the face of delocalization. Death is not annihilation but relational saturation, where the differences of the pattern become indistinguishable from the background of the environment.

Value as difference, opposing delocalisation. If "good" had an ontological correlate (a very big claim), it might be the preservation of structured informational differences against entropic flattening. Art, memory, knowledge, love are all dense local informational structures that resist dissolution.

No cosmic telos, but local significance. At the most basic level, the universe has no “point” – information just gets re-arranged. But for nodes like us, in the network, the experience of building, maintaining and transmitting informational structures is the source of meaning.

These are not arguments but attitudes which our metaphysics makes consistent. We mention them only to show that, unlike some eliminative physicalisms, informational monism does not make room for a humanly meaningful world as a concession but as a consequence of taking emergence seriously.

13. The Problem of Free Will

There is no metaphysical scheme that can avoid the problem of free will. The view of informational monism is non-standard and unique. The deterministic menace. So if the universe is a pure network with fixed rewriting rules R , does this not imply hard determinism? Yes and no. The global evolution is in fact deterministic

in the sense that if you know the entire state of the network at one logical step, there is a unique next step determined by R.

The compatibilist escape hatch. However, fundamental determinism does not imply the non-existence of free will for local subsystems. Free will, to us, is not the ability to break the laws of physics (an incoherent concept), but the experienced ability of a high-density informational node to update its internal state in the face of multiple possible futures, limited by its memory and goals.

Our framework allows this in three ways:

- Informational lack of completeness. No node has ever had access to the full network. So it takes several possible futures (branching) even though only one actually materializes globally. This node really branches.
- Self-modification Unlike classical billiard balls, informational nodes can change their own internal rewriting rules (meta-rules) with time. This is not indeterminism but self-determination — the node becomes part of the causal chain that determines its future states
- Topological freedoms. High-relational-bandwidth nodes have a greater freedom in how they respond to inputs. In this sense, freedom is not the absence of causality but the abundance of responses.

Free will conclusion: informational monism is compatible with a strong compatibilism. Free will is not an illusion , but the experience of being a node of high information density with multiple viable pathways . Just because a Laplacean demon (if such a thing existed) could predict your choice, does not make your deliberation any less real. It just means that the demon has a perspective that you do not share.

14.Synoptic Table: Informational Monism versus rival metaphysics

Feature	Newton	Einstein (GR)	Barbour	Smolin	Our View
Time fundamental	Yes	Yes (as block)	No	Yes	No (emergent)
Space fundamental	Yes	Yes (dynamic)	No	Yes	No (emergent)
Matter fundamental	Yes	Yes	No	Yes	No (information)
Observer dependence	No	No (except SR)	No	No	Yes (time only)
God / Great Observer	Absolute time = divine sensorium	Block = timeless God	Not addressed	Not addressed	Network totality (Spinoza)
Free will	Incompatibilist	Incompatibilist	Illusion	Real (cosmological)	Compatibilist

15.Summary

Informational monism dissolves the aporias of geometric substantialism by refusing to hypostatize space and time. The realization that reality is a mere graph of mathematical relations opens up the possibility to re-unify the phenomenological experience of temporal irreversibility with the causal rigidity of quantum gravity.

Informational monism is not just a re-description of the world, but a real ontological economy: no more substantial spacetime, no more absolute time, no more primitive matter. What is left is a pure network of actualizing differences. Our job is not to fill in this framework, but to learn to think without the categories classical physics has bequeathed to us. ¹

One final note: this essay proposes a metaphysics, not a physics. It does not aim at empirical prediction (although it does some), but at conceptual coherence. In an age when physics is multiplying entities (strings, branes, multiverses) there is intellectual virtue in asking: what is the minimal ontology needed to explain all phenomena? Our answer: pure informational relation. That might be wrong. But it's a question worth asking.

NOTES

¹ This metaphysics is compatible with Spinoza's God—the immanent, timeless totality of all relations—but not with the Abrahamic God as a personal agent acting in time. This is either a virtue or a cost depending on prior commitments.

Glossary

Term	Definition
Ontic information	Pure relational difference constituting the ultimate substrate of reality; not to be confused with epistemic (Shannon) information.
Topological delocalization	Process by which structured information migrates from macroscopic (accessible) to microscopic (inaccessible) degrees of freedom, producing the appearance of increasing entropy.

Term	Definition
Great Observer	Hypothetical omniscient being knowing the complete state of the informational network; a logical limit rather than a possible entity.
Relational bandwidth	Number of distinct connections between two nodes in the graph; measure of informational exchange capacity.
Thermal time	Emergent time defined by a subsystem's thermal state rather than by a fundamental coordinate (Rovelli).
Informational monism	Metaphysical thesis that relational information is the only ultimate substance.

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